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LA-UR-26-28558

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Title: Guidance on Transmission Siting in the United States Transmission Siting Roadmap for the Chiefs' Trail Project

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Intended for: Report

Issued: 2026-09-24



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Guidance on Transmission Siting in the United States

Transmission Siting Roadmap for the **Chiefs' Trail** Project

February 3, 2025

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INTRODUCTION

The *Chiefs' Trail*, also known as *Trail of the Chiefs*, is a project proposed by the National Tribal Energy Association (NTEA): connecting thirteen Native American Reservations across the Western United States with the Western Electricity Coordinating Council (WECC) bulk power system through a new electric transmission corridor. This initiative promotes energy self-determination and economic development among participating tribal communities by allowing them to develop their own energy portfolios and participate in energy markets. The proposed transmission corridor is shown in Figure 1.

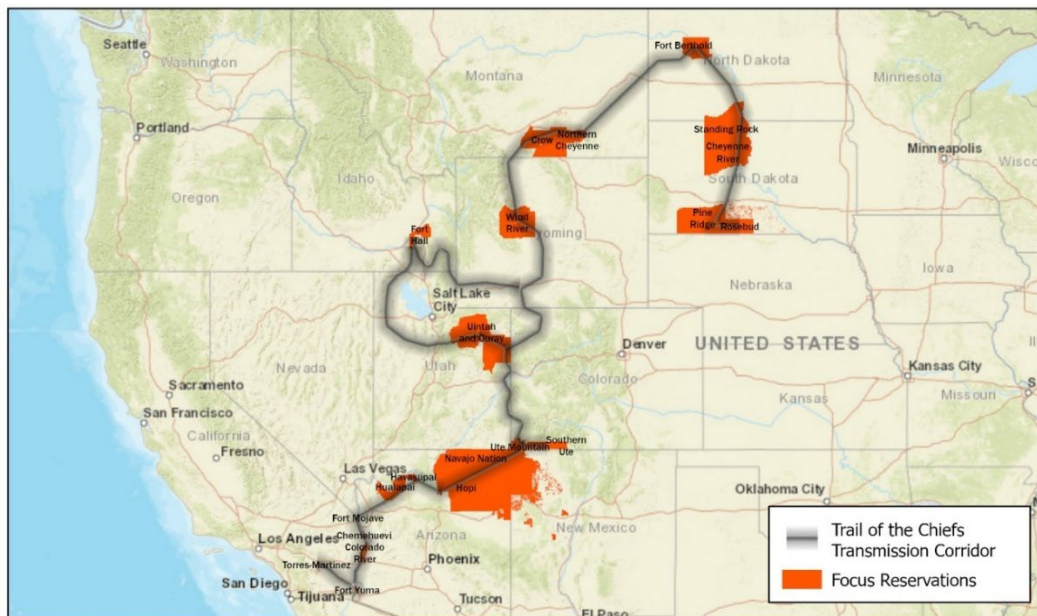


Figure 1. Proposed *Chiefs' Trail* transmission corridor.

The *Chiefs' Trail* will allow Native American Reservations to connect to existing transmission infrastructure, therefore, being able to provide various energy services: alleviating congestion and increasing WECC reliability with a constructed transmission corridor; enabling participation in energy markets using their diverse energy generation portfolios, including the commissioning of new energy generation sites; and providing more affordable electricity to millions of people across the region.

This document serves as a guide for a *Chiefs' Trail* phased-approach feasibility plan, providing a roadmap for transmission siting processes that NTEA could follow, to enable the successful accomplishment of project objectives. The roadmap outlines key considerations, best practices, and regulatory requirements for identifying suitable transmission line routes and integrating new energy generation sites in tribal communities into the WECC system.

A generalized step-by-step approach was designed to help NTEA navigate the complex permitting and approval processes of state and federal agencies—necessary procedures and protocols for engaging with key stakeholders were outlined, which depend on federal- and state-level public utility commissions, and environmental and other regulatory agencies. Additionally, the first step in developing and presenting compelling cases to secure funding opportunities—through the Department of Energy (DOE), the Federal Energy Regulatory Commission (FERC), or other relevant agencies—is provided here, ensuring that tribal communities are well-equipped to effectively communicate the value of the *Chiefs' Trail* and prepared to begin its implementation.

BACKGROUND: TRANSMISSION SITING IN THE UNITED STATES

The term “transmission siting” refers to the process of identifying and evaluating potential locations for new electric transmission lines, towers, and/or substations—facilitating the transmission of electricity from power plants to consumers through the transmission infrastructure.

Transmission siting is critical to ensure the reliable and cost-effective transmission of electricity, as it enables the construction of new routes (while balancing grid reliability, public safety, project costs, and environmental concerns), therefore, allowing remote communities to connect to the national bulk electric system. A transmission siting process typically involves the analyses of land use, zoning regulations, environmental impact, visual and noise impacts, community engagement, and alternative routing options to identify suitable locations and develop proposals for construction or modification of existing infrastructure.

In general, siting new transmission lines is considered one of the most difficult but most important processes of the electric power industry. It involves multiple stakeholders (including federal and state regulators, electric utilities, local communities, and environmental groups) who hold jurisdiction over who, what, and where can build, and who pays for it. It is a complex process, which has evolved over time, but is still considered to be lengthy as it may entail disputes between interested parties at times.

State vs. Federal Jurisdiction

The authority for transmission siting resides primarily with the states. The 2005 Energy Policy Act [1] established a limited federal role, where siting authority is shared between the DOE and FERC. The 2018 FERC Federal Power Act [2], amended in 2021, gives DOE the power to study transmission congestion to provide authorization for designating National Interest Electric Transmission Corridors, commonly known as “national corridors”. This Act grants FERC the authority to issue permits to construct or modify transmission lines and facilities within a national corridor if one of the following statements is true:

1. The state(s) do not have the authority to approve the transmission siting project or do not consider interstate or interregional benefits.
2. The interested entity (e.g., transmission utility) does not qualify for siting approval in a particular state.
3. The state(s) have not decided on the application within 1 year.
4. The state(s) have conditioned the approval of the project such that it will not significantly reduce transmission capacity constraints or congestion or makes it economically infeasible.
5. The state(s) deny the application.

Another point of contention between state and federal jurisdictions is related to electricity rates. For instance, the federal government has jurisdiction over wholesale electricity rates, while states have jurisdiction over retail rates. This dichotomy could create a challenge for transmission siting, as it affects both wholesale and retail aspects of electricity sales due to its effect in congestion and capacity constraints.

State-level public utility commissions (PUCs) have jurisdiction over intrastate transmission lines and must approve or deny requests to construct new ones within their respective states. PUCs consider factors such as need for the transmission line, impact on local communities and environment, and consistency with state energy policies. State regulators determine whether the construction plan of the transmission siting project is done properly and adequately; if considered inadequate, they can forbid the electric utility from charging the costs associated with the project to customers.

Other Considerations

Transmission siting often involves public engagement and participation. Local communities may have concerns about the impact of a new transmission corridor and equipment on their environment, health, quality of life, and energy rates. FERC and PUCs typically provide opportunities for public comment and participation in the transmission siting process by having representatives from the community.

Transmission siting often involves environmental considerations, such as potential impacts on wildlife habitats, water quality, and air quality. Both FERC and PUCs must consider these factors when evaluating a request for construction of a new transmission corridor. Federal and state environmental regulators may become involved in the transmission siting process to address environmental issues arising during a project. The National Environmental Policy Act (NEPA) requires federal agencies to consider the environmental impacts of their actions, including the siting of new transmission lines. State health officials may also become involved in populated areas to assess the effects caused by the electromagnetic fields produced by the current flowing through the transmission lines proposed.

Court Challenges

Transmission siting decisions can be challenged in court by affected parties (including local communities, environmental groups, and electric utilities). These challenges can slow down the transmission siting process and create uncertainty for developers and stakeholders.

Generic Transmission Siting Process

I. Planning: Developers and/or electric utilities perform planning studies to understand the choices they have for delivering reliable and cost-effective electricity to customers. These choices include the construction of new power plants, addition of distributed generation, constructing new transmission lines, implementing demand response programs, developing aggressive energy efficiency programs, or a combination of these. Planners focus on using the proposed transmission infrastructure to reliably deliver power from locations that may have excess capacity to areas with reduced capacity.

II. Cost Analysis: Cost is one of the most important aspects that developers and planners must consider in a transmission siting process—building and commissioning a transmission corridor is a very expensive undertaking. Typical capital cost ranges from \$300K per mile to over \$2M per mile, depending heavily on the voltage class (e.g., 765 kV, 345 kV), cable placement (underground or overhead), and other costs related to land acquisition and the permitting process. Terrain, related to land acquisition, is also a factor that affects cost; for instance, it is more difficult to maneuver construction equipment through mountainous terrain, far from roads, than it is on flat plains with nearby roads. Other factors affecting capital cost include the cost to acquire rights-of-way, the cost of upgrading existing equipment (such as substations), and the cost of interconnecting the new transmission project with the existing grid.

III. Route Evaluation: Planners must perform studies and evaluate multiple possible routes for a transmission corridor proposed. Local interested groups may become involved in a project and may advocate for a particular transmission line route over another. Some issues planners may encounter are related to environmental considerations and the “not in my backyard” syndrome [3], where local opposing groups focus on potential negative impacts a project may cause in their communities, considering recreational land, scenic and historic trails, and parks.

IV. State and Federal Permitting: Developers and/or electric utilities (or simply petitioners) seek permission from state and federal agencies to continue with their transmission siting project. As previously discussed, each state has its own process for granting permission to construct and commission a new transmission line; these processes need to be verified by the federal government. A generic step-by-step description of this follows:

1. Petitioners file an application with the state agency(ies) or local government(s).
2. Once the application is complete, the state agency is given a period to review and rule on the application, which period can last between three months to two years or more (depending on state).
3. Some states require a “certificate of need,” which clearly describes the specific need for the new transmission project; reliability and congestion are “needs” traditionally used for these cases.
4. Some states require a certain number of public hearings to be held that should encompass the affected towns, cities, and counties.
5. Some states require the application to be filed with different alternative routes with an analysis done for each one of them.
6. A procedure is presented to appeal a decision.

V. Financing: Developers and/or electric utilities need to obtain financing for their transmission siting project—this is not a simple task as financing is significant challenge in the transmission sector. One of the reasons why building a new transmission corridor is considered a difficult and time-consuming process is attributed to difficulties in raising capital for large investments, particularly due to the major changes in electric power markets and the financial uncertainties existing in those markets. Higher returns on transmission rates to attract more capital have been proposed, nonetheless, bulk energy rates are still regulated by FERC while retail rates are regulated by the states.

Regulators are exploring solutions to address existing financing issues. One option is the use of performance-based ratemaking plans, where regulators set performance goals and targets, and if those are met, then electric utilities would earn higher returns. This approach aims to incentivize electric utilities to invest in new transmission infrastructure while ensuring that customers pay only for necessary costs [4]. However, it remains unclear who will ultimately bear the costs associated with transmission siting, including regulatory approvals and permitting processes. In some cases, these costs are borne by the developer and/or electric utility itself, while in others they may be passed on to consumers through increased rates. In any case, financing is a critical part of any proposal, which needs to be addressed before the start of a project.

TRANSMISSION SITING ROADMAP

This roadmap describes the transmission siting process for a section of the *Chiefs’ Trail*: the segment of the proposed transmission corridor connecting the Crow and Northern Cheyenne Reservations (Montana) with the Fort Berthold Reservation (North Dakota), shown in Figure 2, named *Montana Chiefs’ Trail*. The focus will be on the permitting and siting processes in the State of Montana—the processes in the State of North Dakota and other states along the *Chiefs’ Trail* will be very similar, with differences in the specific state regulatory agencies involved.

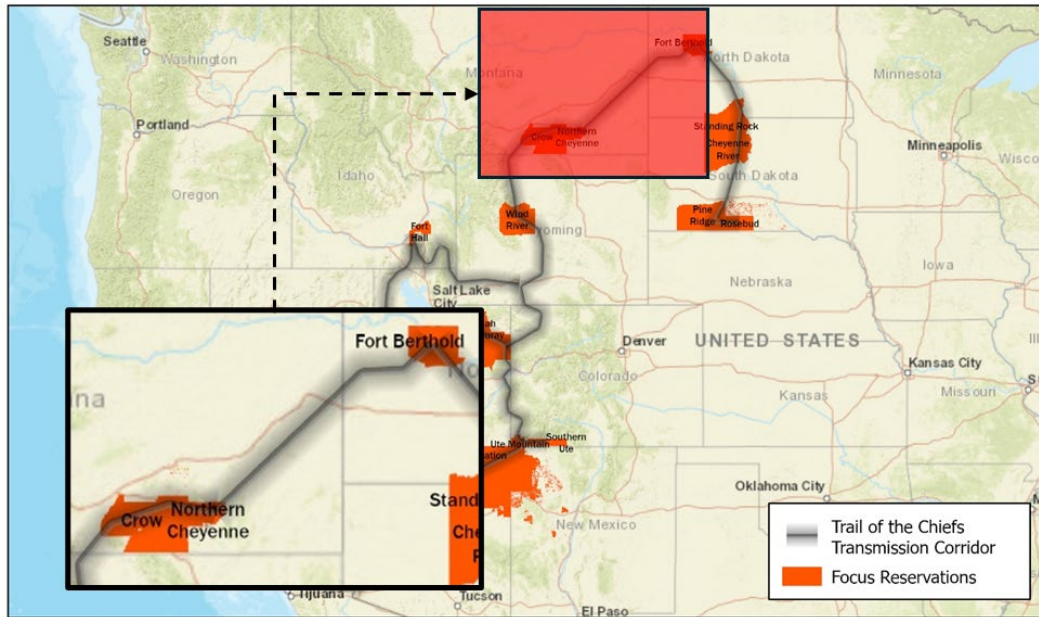


Figure 2. Montana Chiefs' Trail segment for the transmission siting roadmap.

The project timeline of a general transmission siting project is shown in Figure 3. This roadmap is largely based on the timeline proposed by the DOE Regulatory and Permitting Information Desktop (RAPID) toolkit [5] available at <https://openei.org/wiki/RAPID>.

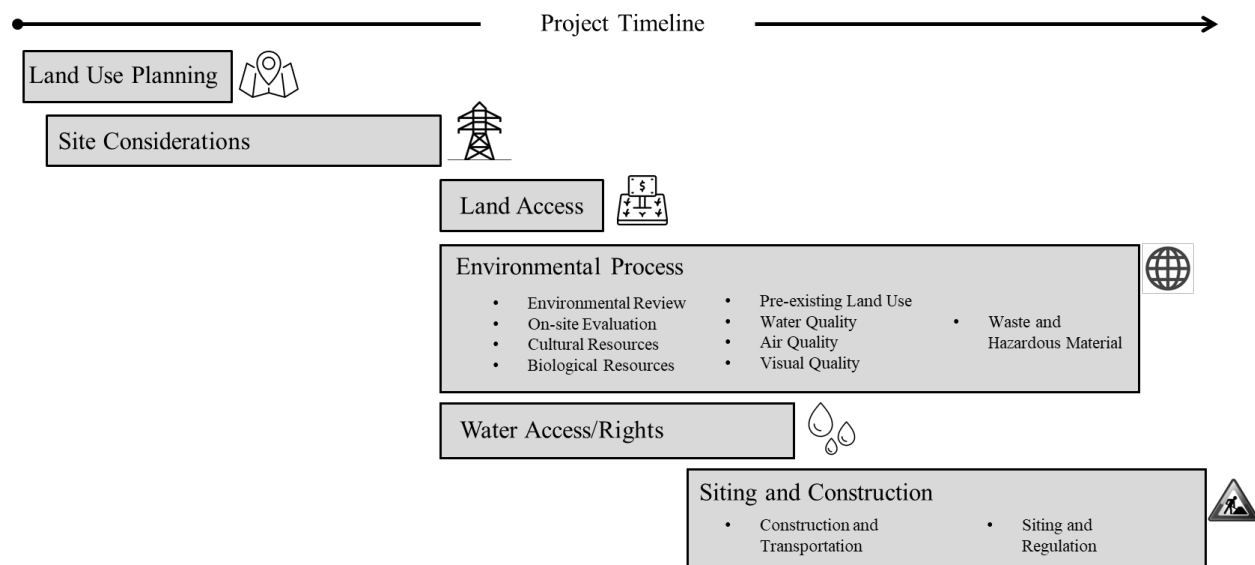


Figure 3. Project timeline of a transmission siting project.

The project development timeline is divided into six major steps: (1) Land of Use Planning; (2) Site Considerations; (3) Land Access; (4) Environmental Process; (5) Water Access and Rights; and (6) Siting and Construction Process.

Preliminary Stage

Developers should start the planning with understanding the context of their proposed transmission siting project. For example, identifying the interconnection(s) and the North American Electric Reliability Corporation (NERC) regulated region(s) their state is connected to, the existing transmission owners and operators involved, and the other agencies that may become part of the permitting and siting processes.

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Consider before proceeding:

- 1) The electrical grid is part of both the Western and the Eastern Interconnections.
- 2) The electrical grid is part of two NERC regulated regions, the Midwest Reliability Organization (MRO) [6] and the Western Electricity Coordinating Council (WECC) [7], who provide regulations for operating and planning activities. WECC is the Regional Entity responsible for coordinating bulk power system reliability in the Western Interconnection; MRO serves a similar function within its service territory in the Eastern Interconnection.
- 3) Certain transmission owners are part of the Northern Tier Transmission Group, a group of providers and customers who are actively involved in the sale and purchase of transmission capacity, supplying electricity to customers in the Northwest and Mountain States.
- 4) There are a few investor-owned, independent, and municipal transmission providers, such as:
 - a. Central Montana E Pwr Coop Inc, serving rural electric utilities across Montana;
 - b. Montana-Dakota Utilities, serving Eastern Montana; and
 - c. Northwestern Energy, serving Western Montana.
- 5) The Montana Department of Commerce (DOC) Energy Promotion and Development (EPD) Division, created in 2007, promotes responsible energy development of Montana's resources. EPD works alongside the DOC to facilitate various processes (including permitting, siting, workforce training, and financial assistance), with a focus on developing transmission and infrastructure for Montana's energy resources.
- 6) Throughout the proposed project, NTEA and the tribal communities involved must actively participate in the decision-making processes.

Step 1: Land Use Planning

The first step—which should happen concurrently with **Step 2**—is developers ensuring that their proposed transmission siting project complies with existing federal, state, and local land use plans and zoning regulations. Federal and states agencies use Land Use Plans (LUPs) to manage the use of their lands and dictate what can and cannot be done in the land. In addition to LUPs, state and local governments have zoning regulations that may restrict development activities: dictate building types, transmission line locations, setbacks from streets, and other boundaries (such as the size and height of buildings).

States often delegate land use planning and zoning oversight to local governments (such as counties and municipalities), with some states having a cooperative process between state and local governments. Level of control over land use planning varies by state, ranging from statewide plans to local government control.

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In Montana, LUPs are delegated primarily to local governments pursuant to the Local Planning Enabling Act (M.C.A. secs. 71-1-01 to 71-1-606). Local governments are given authority to regulate and restrict the use of land and require permits for any construction project within their jurisdiction. The Montana Municipal Zoning Enabling Act (Montana Code 76-2-301, 302) [8] allows cities and towns to control zoning and building, while the County Planning and Zoning Commission Act (Montana Code 76-2-201 [9]) enables counties to adopt their own zoning regulations. Any transmission siting project will need to obtain land use permits from each local government involved, but the specific regulations may differ depending on the local jurisdiction.

Developers must comply with regulations established by the Bureau of Land Management (BLM), who is responsible for managing federal natural resources and sustaining the health, diversity, and productivity of public lands. BLM is in charge of roughly 258 million acres of public lands and 700 million acres of mineral estates. They develop and maintain their own LUPs, called Resource Management Plans (RMPs), which outline areas with limited/restricted/exclusive use, and designate areas of critical environmental concern where development actions are prohibited.

A transmission siting project that passes through BLM-managed lands requires environmental review processes, which include compliance with specific RMPs and with the National Environmental Policy Act (NEPA). NEPA is a federal law, enacted in 1970, which requires federal agencies to assess the potential environmental impacts of major federal actions before making decisions.

Another review that developers must complete checks whether the transmission siting project is proposed to be constructed on land managed by the United States Forest Service (USFS). USFS manages approximately 192 million acres of land in accordance with the requirements of the 1976 National Forest Management Act. USFS must also develop LUPs, called Land Management Plans (LMPs), that are prepared and implemented individually by each national forest. Transmission siting projects are required to comply with specific forest LMP(s).

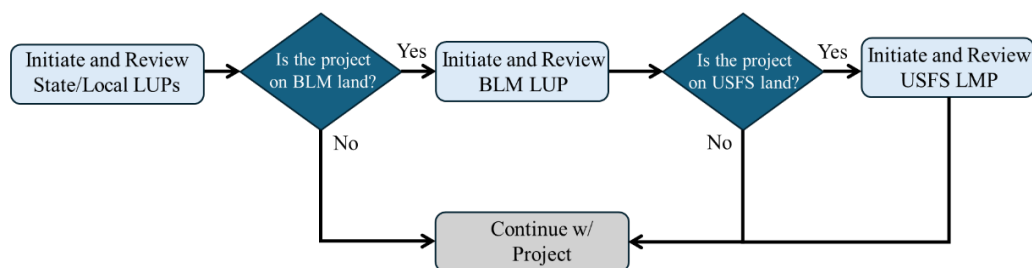


Figure 4. Overview of the Land Use Planning processes.

The flow diagram of Figure 4 presents the **Land Use Planning** processes. For each review process (i.e., BLM and/or USFS), plan revisions or amendments may be required, and an Environmental Impact Statement (EIS) should be prepared by BLM.

Step 2: Site Considerations

The second step is considered to be the most technical one as developers must perform site selection analyses and studies that involve factors like:

- pre-existing uses of the land, which is proposed to be used for the project;
- land access requirements;
- proximity to existing transmission lines and rights-of-way;
- proximity to existing land and highway rights-of-way;
- transmission siting and interconnection requirements;
- pre-existing recreational resources near the proposed project site; and
- cultural and environmental resources near the proposed project site that may be impacted (i.e., cultural resources, water resources, ecological resources, geological resources, biological resources).

2.1 Pre-existing Land Restrictions

Developers must consider whether the location of their proposed transmission siting project has pre-existing land use restrictions, particularly if that land is managed by a federal or state agency; certain lands have statutory or regulatory limitations (such as federally designated wilderness areas and primitive areas managed by USFS). Before making substantial investments, developers should research whether a proposed site is available for development, and review all guidance from federal agencies, including the Bureau of Reclamation (BOR), USFS, and other agencies that may have stipulations in place to protect environmental resources. Restricted areas include national parks, national monuments, recreational areas, national wildlife refuges, fish hatcheries, wild and scenic rivers, Native American Reservations, historical/cultural centers, areas with threatened and endangered species, military areas, and water conservation areas.

Federal agencies are required by Executive Order 11988 [10] to minimize the risk of flood loss and protect human safety, health, and welfare in floodplain areas. If a proposed transmission siting project is located within a floodplain, agencies must take specific actions during the environmental review under NEPA, including considering alternatives that avoid adverse effects and incompatible development in these sensitive areas. This review process should go hand in hand with **Step 1**.

2.2 Land Access Requirements and Proximity to Existing Rights-of-Way

When selecting the location for their transmission siting project, developers must consider two key factors: (1) land access requirements, and (2) proximity to existing rights-of-way. A project will likely require access to multiple types of lands (including federal, state, local, tribal, and private property), which necessitates to obtain rights-of-way, leases, or use authorizations, or purchase the land outright, depending on land ownership status.

2.3 Siting and Interconnection Requirements

Developers must consider the site location's proximity to existing transmission lines and interconnection(s), and what approvals will be required to continue with the project. During this process, the siting team needs to work with system planners.

The flow diagram of Figure 5 presents the *Transmission Siting* and *Interconnection* processes: **System Planning Studies** are followed by **Siting Studies**, and only when those are finalized developers can go into the **Approvals** review process (which may send back the entire process to the planning stage) [11].

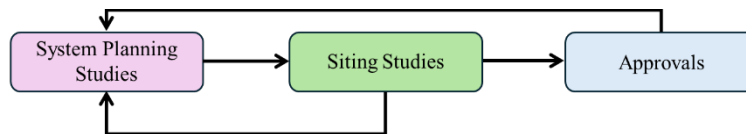


Figure 5. Overview of the *Transmission Siting and Interconnection* processes.

System Planning Studies

Developers are required to justify the need for the transmission siting project, identify alternatives for the interconnection, and select the voltage level(s) and terminal locations of the proposed transmission line(s). *Power flow* studies are the foundation of every study during the planning phase of the project [11].

Transmission System Planning Studies provide the technical foundation for siting a transmission line(s), and guide project decisions from start to finish. These studies determine the optimal voltage level(s) and terminal locations for new line(s), ensuring reliable power flows to loads at minimal cost. The goal is to create a resilient power grid interconnection capable of withstanding disruptions. While planners drive the planning process, they do not oversee the physical design, operation, or maintenance of a line(s). Planners focus on running a series of electrical studies (i.e., power flows) to analyze the transmission network as it currently exists, and then evaluate the effect that the new transmission line(s) or whole project will have. They evaluate multiple generation and loading scenarios (e.g., worst case peak loading) with computer simulations, which use models capable of predicting the operation of the bulk power system during the simulated conditions.

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I. Transmission Line Interconnection Feasibility Study: The technical viability of constructing and integrating the proposed transmission corridor into the Western and the Eastern Interconnections (managed by WECC and MRO, respectively) must be assessed with computer simulations and power flow studies. These studies need to address grid stability, congestion, voltage, and other potential technical issues that could arise from the construction and integration of new transmission infrastructure. At this step, system planners must select the voltage for the new transmission line(s) and facilities, and need to make assumptions about other details such as ampacity and conductor size of the line(s).

II. Loading Analysis: Detailed analyses of the current and the future electricity demands of each Native American Reservation must be conducted to understand their energy needs, energy consumption patterns, anticipated growth, and the potential energy markets they could participate within the WECC and/or MRO. At this step, historical electricity consumption data from the Reservations must be gathered, and load profiles generated based on these data to understand their energy usage patterns; load information from the WECC and the MRO systems also need to be obtained to model the entire representative bulk power system. Discussions with tribal representatives will be needed to understand and project their future energy needs based on factors such as population growth, economic development, and electrification efforts.

III. Energy Resource Assessment: An assessment of energy resources available within the tribal territories must be conducted. This study needs to be done using information on existing generation resources that tribes have access to and are planning to connect to the bulk power system through the proposed transmission corridor—furthermore, the technical viability of integrating new energy generation sites needs to be assessed. A full *Interconnection* process, somewhat similar but separate from the *Transmission Siting* process, will be necessary.

New generation sites are planned to be managed by individual tribal communities, considering energy resource potential within their own territories. The assessment should evaluate the potential for connecting

and siting new and existing traditional, solar, wind, geothermal, and biomass sites, focusing on how these resources can be used to supply energy across the *Chiefs' Trail*, and how these will allow participating communities to sell energy back to the electric grid.

Siting Studies

In the **Transmission Design** process, developers must select companies specialized in designing and constructing overhead or underground transmission lines. The three main components that need to be designed are the wires (conductor and shield wire), the structure, and the foundation [11]; this is a complex task as transmission lines must meet adequate electrical, strength, and aesthetic requirements.

For wire design, the conductor wires need to meet the requirements set by the **System Planning Studies** and the design standards set by the electric utility. The choice of conductor type and size will determine electrical capacity, electrical losses, thermal capacity, and corona noise—it must be noted that this selection will directly affect the total capital cost of the transmission siting project as it will determine the structure configuration and spacing of the new transmission infrastructure. Mechanical and electrical properties (such as wire sag, tension, and clearance) must be taken into consideration during the design.

For structure design, an appropriate structure for the poles or towers, which will support the transmission line(s), needs to be selected. Structure configuration is influenced by the **System Planning Studies** requirements, siting recommendations, geography, utility standards, land requirements, and permitting requirements. Some of the most used structures for transmission lines are poles or lattice towers made up of various materials (such as wood, concrete, or steel)—**Figure 6** presents commonly used structure designs.

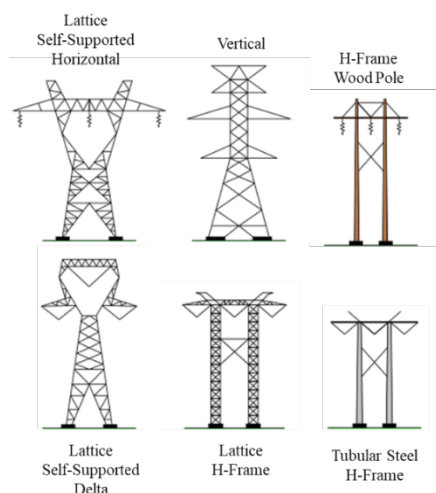


Figure 6. Transmission line structure designs [12].

For foundation design, the decision typically occurs near the end of the project's final design phase, following a thorough sub-surface investigation; however, it may be necessary to make the decision earlier to meet regulatory requirements. Foundations can range from simple direct burial systems to more complex designs (such as reinforced concrete drilled shafts), which are among the most commonly used ones in the United States—other examples include grillage, driven piles, vibratory foundations, and micro-piles [11].

It is beyond the scope of this document to provide a comprehensive overview of the large number of decisions involved in a **Transmission Design** process. Designed transmission lines should meet specific project requirements and must satisfy all applicable design codes, the most important being the latest National

Electric Safety Code standard that dictates minimum design requirements—however, most utilities and regulators have more stringent design criteria of their own.

In the *Transmission Line Route Analysis and Assessment* process, developers must evaluate potential transmission line routes that can connect the new transmission infrastructure with the bulk power system. First, analysts determine the area that needs to be studied for the transmission siting project; this must be large enough to encompass all possible alternative routes that should be considered. Next, the site of the substation(s) needed for the project must be selected—the siting of a new substation(s) has significant impact on transmission line routing options, which in turn affects its approval. Careful consideration should be given to the selection of the substation site(s) to ensure that they align not only with regulatory and environmental requirements, but also comply with engineering factors as well; the location of the substation(s) ensures that the system can operate as planned in the **System Planning Studies**.

In parallel to the substation site selection, an expert team needs to identify, evaluate, and determine the route of the proposed transmission line(s). These routes should, as much as possible, minimize: route length and cost; interference with existing residential and non-residential structures, historical/cultural or recreational facilities, public resource lands (such as forests and parks), and economic activities; and high-visual impact on residential or public areas. To perform these studies and develop alternative routes, Geographic Information System (GIS) data needs to be collected; software can be used to map the proposed transmission siting project to help decision-making. Multiple alternative routes must be identified and refined based on routing constraints (identified using collected data) and public involvement. **Figure 7** shows a generic method that determines how potential routes of a network can be defined between substations.

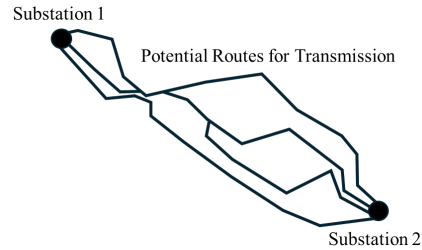


Figure 7. Potential network routes for transmission line siting.

Existing transmission corridors offer significant opportunities for optimizing new transmission line routes. Various federal and state regulatory agencies have shown a strong preference for utilizing existing corridors when constructing new transmission infrastructure. By taking advantage of these corridors, projects can minimize potential opposition and ensure smoother regulatory approvals [11].

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Potential transmission line routes, which can connect all three Native American Reservations to the WECC and/or MRO systems, need to be identified. Both overhead and underground transmission lines (with both AC and DC options) need to be assessed, considering key factors such as geographic obstacles and land ownership. GIS and geospatial data need to be used to identify potential obstacles such as protected areas, water bodies, and sensitive habitats. Optimal connection points to the WECC and/or MRO systems need to be evaluated, considering existing transmission infrastructure and its corresponding capacity derived from models.

Approvals

Any new transmission infrastructure related to the proposed transmission siting project needs approvals (from federal, state, and/or local authorities) before construction can begin. It is usually in this phase of a project when most proposed projects fail to be realized—this is why it is important to obtain all necessary approvals before continuing with the project. Some of the common questions developers may use as guidance when obtaining approvals [5]:

1. Does the transmission siting project require federal authorizations?
2. Is the project proposed in a National Interest Electric Transmission Corridor?
3. Do the state(s) have a comprehensive transmission siting process?
4. Does the project require approval from a state utility regulatory authority?
5. Does the project require any additional approvals?
6. Does the project require local approvals?

Figure 8 shows a simplified flowchart of the general approval process.

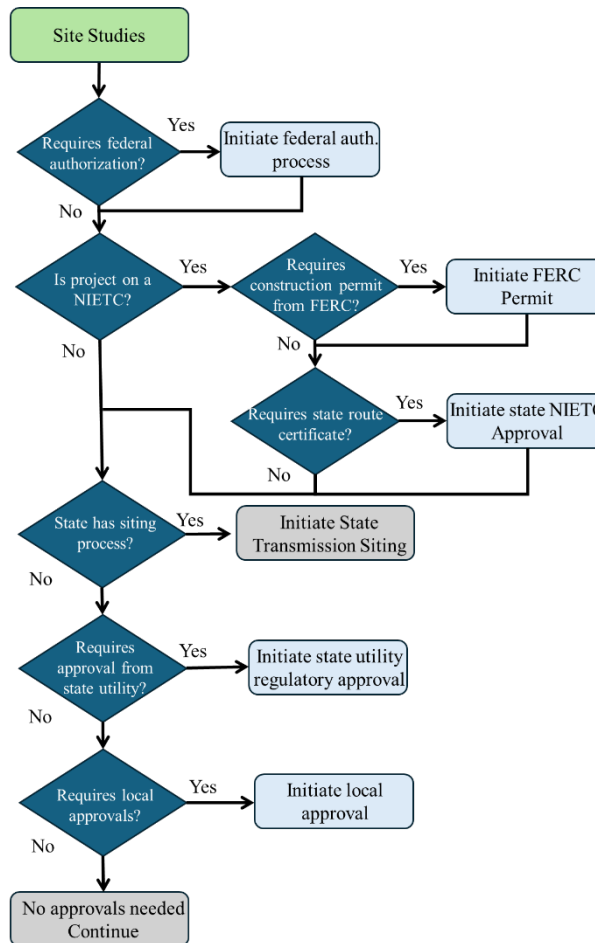


Figure 8. Flowchart for approvals and/or permits needed for a transmission siting project.

Montana Chiefs' Trail

In Montana, high-voltage transmission line siting is managed by the Montana Department of Environmental Quality (MDEQ). The state-administered siting act, called Montana Major Facility Siting Act (MFSA) (Montana Code 75-20 [13]), requires that developers file an application, called Certificate of Compliance (CoC), with MDEQ who has the authority to approve or deny their request; Montana law requires that all proposed siting projects with a voltage over 69 kV must obtain a CoC.

The definition of transmission infrastructure does not include (per Montana Code 75-20-104 [14]):

- transmission lines of < 230 kV and ≤ 10 miles in length;
- transmission lines of > 69kV for which right-of-way agreements have been reached with more than 75% of landowners who collectively own more than 75% of the property along the right-of-way;
- transmission lines that are < 150 miles in length and are required under state or federal regulations (due to reliability, interconnection to regional transmission grid, or secure firm transmission service);
- upgrades to an existing transmission line of a design capacity of > 50 kV to increase capacity; and
- transmission substations, switchyards, or other control equipment facilities (including energy storage facilities).

A CoC application contains three sections: (1) *Alternative Siting Study*, (2) *Baseline Study*, and (3) *Impact Assessment*—MDEQ provides detailed requirements for these in their "Application Requirements for

Linear Facilities" circular. The transmission siting process also involves public meetings and an overview of public issues and concerns related to each alternative transmission line route.

MDEQ reviews and notifies an applicant of the completeness of their CoC application and whether revisions are necessary within 30 days of the submission. Nine months after a complete application is submitted, MDEQ issues a report detailing alternatives, environmental findings, and opinions; the general format of the report is an environmental impact statement. Once the proposed transmission siting project is approved through the CoC, the applicant can use eminent domain to secure easement rights because the project is deemed a public necessity.

In terms of local requirements, counties and municipalities—per the Montana Municipal Zoning Enabling Act (Montana Code 76-2-301) and the Montana County Planning and Zoning Commission Act (Montana Code 76-2-201)—have zoning authorities to regulate land use and require permits for constructions that lie within their jurisdictions. However, MDEQ can refuse to apply any local laws or regulations that are considered unreasonably restrictive in view of public necessity (Montana Code 75-20-301).

Step 3: Land Access

The third step is developers obtaining approvals (licenses) and/or purchasing the land for constructing the proposed transmission siting project—land access is a critical stage as it determines feasibility and successful implementation. Securing access to land through negotiations with property owners, easements, or rights-of-way agreements—where owners can be private owners, local agencies, or governmental agencies—allows developers to build the transmission infrastructure necessary for realizing their project. Without land access, a project may be delayed or abandoned altogether, resulting in significant financial losses and potential adverse impacts to the long-term reliability of the bulk power system.

Some of the common questions developers may use as guidance:

1. Does the transmission siting project require access to private land?
YES: A review of the private land must be conducted before signing the easement or purchasing the property.
2. Does the transmission siting project require access to local government land?
YES: Developers should contact the local government with jurisdiction over the land.
3. Does the transmission siting project require access to state land?
YES: Developers should obtain a lease, right-of-way, or other approvals from the state agency with jurisdiction over the land.
4. Will the transmission siting project encroach on a local and/or state highway right-of-way?
YES: Developers should obtain a lease, right-of-way, or other approvals from the local and/or state agency with jurisdiction for projects that encroach on a local and/or state highway.
5. Does the transmission siting project require access to tribal land?
YES: Developers should obtain approvals from the respective tribe(s) through Tribal Energy Resource Agreement(s), an authorization from the United States Department of Interior.
6. Does the transmission siting project require access to federal land?
YES: Developers should contact the federal agency in charge of the land, including, but not limited to: BLM, USFS, BOR, National Park Service (NPS), the United States Fish and Wildlife Service (FWS), and the United States Department of Defense (DOD)—each may have different application processes to obtain right-of-way approvals.

7. Is the transmission siting project within the right-of-way of federal aid or direct federal highway projects?

YES: Developers should obtain permission from the Federal Highway Administration (FHWA) to use federal aid highway rights-of-way. Before approval, FHWA (1) ascertains the impacts the project would have on safety, (2) determines environmental and economic impacts on agriculture, and (3) considers environmental and economic impacts the project would have on the use of the highway.

Montana Chiefs' Trail

To obtain land access, developers must work with the Montana Trust Land Management Division (MTLMD), the Montana State Land Board (MSLB), and/or the Montana Department of Natural Resources and Conservation (MDNRC). The Real Estate Management Bureau of MTLMD is in charge of reviewing and processing rights-of-way applications and easements across the trust lands. In case the proposed project requires the use of- or access across state lands, MSLB is the responsible authority to contact. In case the proposed project requires the casual use of state lands prior to the issuance of a full lease, submission of a Land Use License Application to MDNRC is necessary [5].

Additionally, in case the proposed project is projected to have an impact on the state highway rights-of-way, developers must apply for a permit from the Montana Department of Transportation (MDOT), which allows building new encroachments (such as new approaches, facilities, and pole lines)—depending on the type of the encroachment, developers may need to submit an Encroachment Permit Application or an Application for Right of Way Easement.

Step 4: Environmental Review Process

The fourth step is developers ensuring that all appropriate environmental approvals have been obtained before continuing with the proposed transmission siting project. This encompasses the following:

I. Environmental Review: Developers must comply with federal and state environmental review processes. Any environmental issue that requires permitting and/or regulatory approval (such as construction, operation, or maintenance activities) must be addressed before continuing with the project. To initiate a review, developers must determine whether the project meets the “major federal action” criteria, defined as “*an action that the agency carrying out such action determines is subject to substantial federal control and responsibility*” (42 U.S.C. 4336e (10)). If applicable, the project must comply with the NEPA review process—the scope and depth of this will depend on the project and the level of federal involvement; in some cases, projects may qualify for a categorical exclusion, while others may require a comprehensive EIS.

In addition to federal environmental regulations, the proposed project must also comply with state environmental laws, which could include evaluations related to noise disturbances, vehicle emissions, and direct and indirect impacts to cultural resources, ecological resources, and water quality.

Montana Chiefs' Trail

Montana does not have a state-specific environmental review process for transmission siting projects.

II. On-site Evaluation: Developers should conduct on-site evaluations, designed to identify potential land use restrictions and site-specific resource protection concerns, which may impact the proposed project. The

assessment involves consulting with relevant federal, state, and local governmental agencies to determine the necessary regulations to meet and permits to obtain. Multiple factors—including biological, ecological, paleontological, archaeological, geological, hydrological, and aesthetic impacts—can affect the site's usability, which must be considered when performing the evaluation.

III. Cultural Resources: Developers must consider the potential impacts of the proposed project on historical/cultural resources (including artifacts, human remains, historic sites, and locations with cultural importance); federal and state laws protect these. It is essential to conduct an initial assessment to identify any protected resources that may be affected by the project, ensuring compliance.

Montana Chiefs' Trail

Developers should closely coordinate with the Montana State Historic Preservation Office to avoid any unintended disruption to historic sites. In case cultural resources are discovered on a project site, developers must comply with Montana state law, which requires obtaining a permit and providing public notice.

IV. Biological Resources: Developers must consider the potential impacts of the proposed project on biological resources, including flora and fauna, as well as their habitats. Federal and state laws mandate a comprehensive assessment—evaluating the impacts of access road development, transmission line installation, or other related site work—as construction may lead to wildlife habitat losses, reduction in plant diversity, increased erosion, increased dust deposition, and elevated risk of wildfires.

Montana Chiefs' Trail

Developers must consult with the Montana Fish, Wildlife, and Parks Division (MFWPD) to determine whether protected wildlife habitats will be affected by the proposed project. MFWPD will provide measures to avoid or mitigate damage to local ecosystems.

V. Pre-existing Land Use: Developers must assess existing land uses within and surrounding the proposed project site. Federal and state laws may impose restrictions on altering or interfering with certain pre-existing land uses (such as farmland, livestock operations, nearby airports, military bases, navigable waters, and other sensitive areas). For a comprehensive description, see **Step 1: Land Use Planning**.

VI. Water Quality: Developers must evaluate the potential impacts of the proposed project on water quality. Water quality is protected by the Clean Water Act, aimed to reduce water pollution and ensure that surface waters are safe for human consumption, recreation, and aquatic ecosystems. Development activities associated with a transmission siting project can pose significant risks to water quality, including soil erosion (creating increased turbidity and sediment transport), weathering of newly exposed soils (causing leaching and oxidation), discharges of waste or sanitary water, and the use of untreated groundwater (depositing dissolved salts on the surface and entering surface water systems). To mitigate risks, developers must implement measures to prevent water pollution and ensure compliance with federal regulations.

Montana Chiefs' Trail

In Montana, a transmission siting project is required to obtain several water quality permits: a Nonpoint Source Pollution permit, a National Pollutant Discharge Elimination System (NPDES) permit, and a Section 401 Water Quality Certification. Developers must contact MDEQ, who administers the voluntary regulatory Nonpoint Source Pollution Management Program, to check whether their project complies with the program. Developers must also comply with NPDES requirements, a permit granted by the United States Environmental Protection Agency (EPA), if the proposed project requires the discharge of pollutants into water systems (including wetlands). Last, an MDEQ Section 401 Water Quality Certification must be obtained if the project requires federal licenses or permits to construct or operate a facility that may result in any fill or discharge into water systems.

VII. Air Quality: Developers must comply with air quality standards set by federal, state, and local agencies. In the United States, EPA is in charge of administering the Clean Air Act, which empowers states, territories, and tribes to share authority in regulating and providing air quality permits. This Act sets stringent standards to safeguard public health from pollutants (such as carbon monoxide, nitrogen oxides, ozone, particulate matter, and lead), and imposes limitations on both stationary and mobile sources of pollution (such as from vehicles, heavy construction equipment, diesel generators, fugitive dust emissions, blasting activities), requiring the implementation of operational measures that restrict air pollution. Before initiating the proposed project, developers must assess how the project may generate pollution.

Montana Chiefs' Trail

Developers must adhere to specific air quality standards related to particulate matter (PM), visibility, and PM10 emissions, as outlined in Montana Administrative Rules 17.8.220, 17.8.221, and 17.8.223 [15]. Developers must comply with opacity restrictions governed by 17.8.304 [16], which mandates that emissions do not exceed 20% opacity over a 6-minute period. Developers should also be aware of local regulations in Montana's particulate matter nonattainment areas, where consultations with county health departments are recommended to ensure compliance with dust rules.

Under Montana state regulations, the proposed transmission siting project would be generally exempt from requiring an Air Quality Permit from MDEQ Air Quality Bureau [5]. For the construction phase specifically, temporary use of portable diesel generators does not necessitate air quality permits since these generators are classified as nonroad engines under federal regulations, as outlined in 17.8.743(1), 17.8.748(1), and 17.8.744(1)(b).

VIII. Geological Resources: Developers must assess the potential impacts of the proposed project on geological and paleontological resources. These impacts can be caused directly by construction activities (such as excavating, blasting activities, construction of access roads, altering natural drainage, and cutting slopes, or indirectly by soil erosion).

Montana Chiefs' Trail

Developers must consult with the United State Office of Indian Energy and Economic Development, and comply with all federal rules related to geological or paleontological resources.

IX. Aesthetic and Recreational Resources: Developers must consider the potential impacts of the proposed project on visual resources, because transmission lines may cause visual contrast within the surrounding landscape due to their length and size, as well as the regular geometric forms of the transmission towers—this may affect sensitive viewers (such as residents and recreationist) located along the rights-of-way.

During the **Step 1: Land Use Planning** process, developers must assess the project's impact on aesthetic and visual resources to ensure that public views are preserved.

Montana Chiefs' Trail

In the MDEQ CoC application, an assessment of the impacts to visual resources must be included—see **Step 2: Site Considerations – Approvals**. In this assessment, developers should consider how the proposed construction areas may conflict with existing visual management plans or regulations designed to protect scenic views. The CoC *Baseline Study* must include information about potential visual impacts on sensitive areas (such as national parks, residential zones, and highways located within the project's construction zone). Developers must assess the potential types and levels of visual resource impacts for each alternative location proposed, including the potential alteration and impact on affected lands, changes to visual quality, and mitigation measures.

X. Waste and Hazardous Materials: Developers must consider the potential environmental impacts associated with waste and hazardous material generation, disposal, storage, and management. This includes assessing whether the proposed project will generate, store, or dispose waste or hazardous materials (such as underground and aboveground storage tanks, which may trigger additional permit requirements [5]). Construction activities are likely to produce solid and industrial waste, including non-hazardous items (such as containers and packaging materials), as well as hazardous substances (such as fuels, lubricating oils, hydraulic fluids, and compressed gases used for welding and cutting).

Montana Chiefs' Trail

Developers must obtain an MDEQ Hazardous Waste Permit if the proposed project will treat, store, dispose, or transport any hazardous waste. The applicable hazardous waste statutes and regulations, the Montana Hazardous Waste Act and Montana Administrative Rule 17.53, are based on federal regulations: Title 40 CFR 260-270 Hazardous Waste.

Step 5: Water Access Rights

The fifth step is developers assessing the water access and water right needs of the proposed transmission siting project—these are primarily regulated by state law. Depending on project requirements for dust suppression and construction activities, developers may need to secure water from a leaseholder, municipality, or permanent water right holder. Water rights can be divided into groundwater and surface water rights, with different allocation systems in place across the Western and Eastern United States. In the West, it is primarily based on prior appropriation (i.e., first-in-time, first-in-right), while in the East, it is based on riparian rights (i.e., property ownership adjacent to a body of water).

Step 6: Construction and Siting

The sixth and final step is constructing the proposed transmission siting project, as long as all permits and approvals have been obtained. It is beyond the scope of this document to provide an overview of all additional permits and authorizations required for the construction of the proposed project.

STATEMENT OF WORK: MONTANA CHIEFS' TRAIL

This document has described the transmission siting process for the *Montana Chiefs' Trail*. The outlined roadmap for this segment of the proposed *Chiefs' Trail* transmission corridor was designed to help developers navigate the complex permitting and approval processes involved.

In this section, a statement of work (SOW) is introduced: a feasibility plan is proposed, and potential alternatives and next steps are presented, to ensure that NTEA and the tribal communities are well-prepared to pursue funding opportunities for implementation, and well-equipped to continue the planning of other segments of the *Chiefs' Trail* project.

Statement of Work

The flow diagram of **Figure 9**—that expands on the project development timeline of Figure 3—highlights the processes (analyses and studies) in which the Los Alamos National Laboratory (LANL), the DOE North American Energy Resilience Model (NAERM), and the DOE Intermountain West Energy Sustainability and Transitions (I-WEST) teams could provide support.

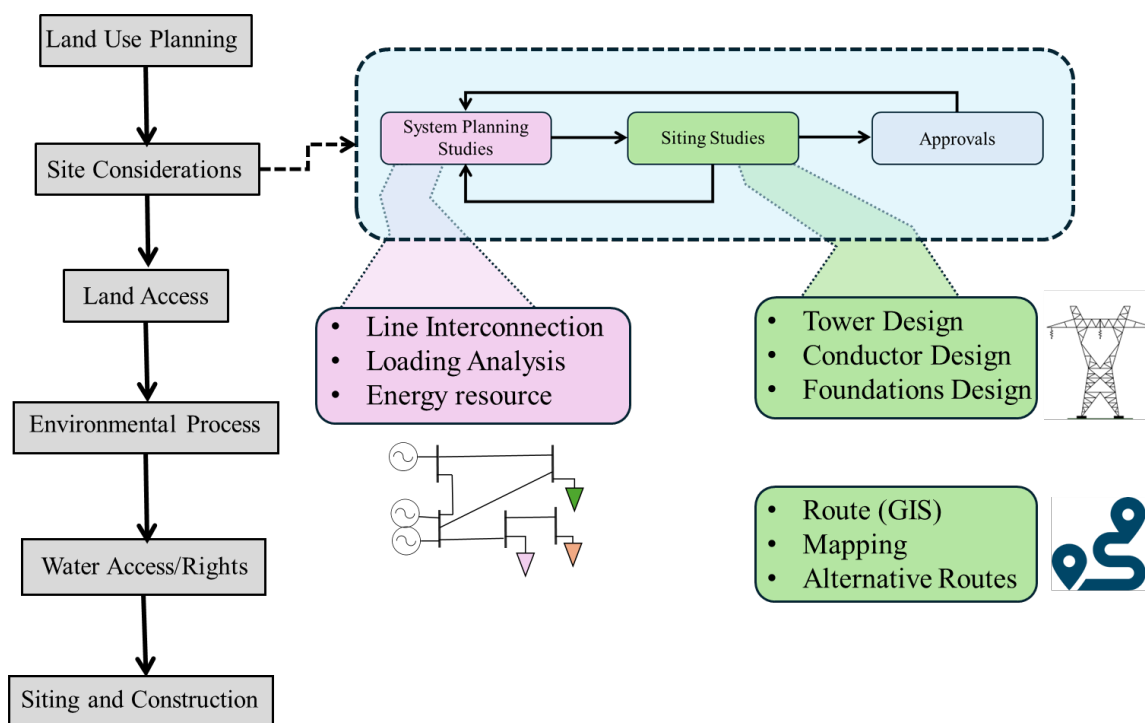


Figure 9. Overview of a transmission siting project, with emphasis on **Step 2: Site Considerations**.

LANL, partnering with the NAERM and the I-WEST teams, has the capabilities to perform the **System Planning Studies** and the **Siting Studies**—specifically, the line interconnection, the loading, and the energy resource analyses, furthermore, the routing and the mapping studies. However, implementation of the *Montana Chiefs' Trail* requires the participation of companies specialized in designing and constructing overhead or underground transmission lines—specifically, to design the foundation, the towers, and the conductors.

SOW for Step 2: Site Considerations should outline the key tasks and objectives necessary to proceed—this could be for the whole *Chiefs’ Trail* or a segment, such as the *Montana Chiefs’ Trail*, depending on the preferences of NTEA.

The specific SOW for the *Montana Chiefs’ Trail* is presented below—SOWs for other segments of the *Chiefs’ Trail* project would be comparable to this one.

MONTANA CHIEFS’ TRAIL				
TASK	PERFORMER	BUDGET	DURATION	
Phase 1 – Transmission Line Interconnection and Route Analysis	NAERM	\$350k	6 months	
Phase 2 – Energy Resource Assessment	NAERM	\$300k	7 months	
Phase 3 – Loading Analysis	NAERM	\$150k	3 months	
Phase 4 – Transmission Infrastructure Design	External	\$300k	6 months	
Phase 5 – Community Engagement	I-WEST	\$200k	4 months	
Phase 6 – Risk Assessment	I-WEST	\$200k	4 months	

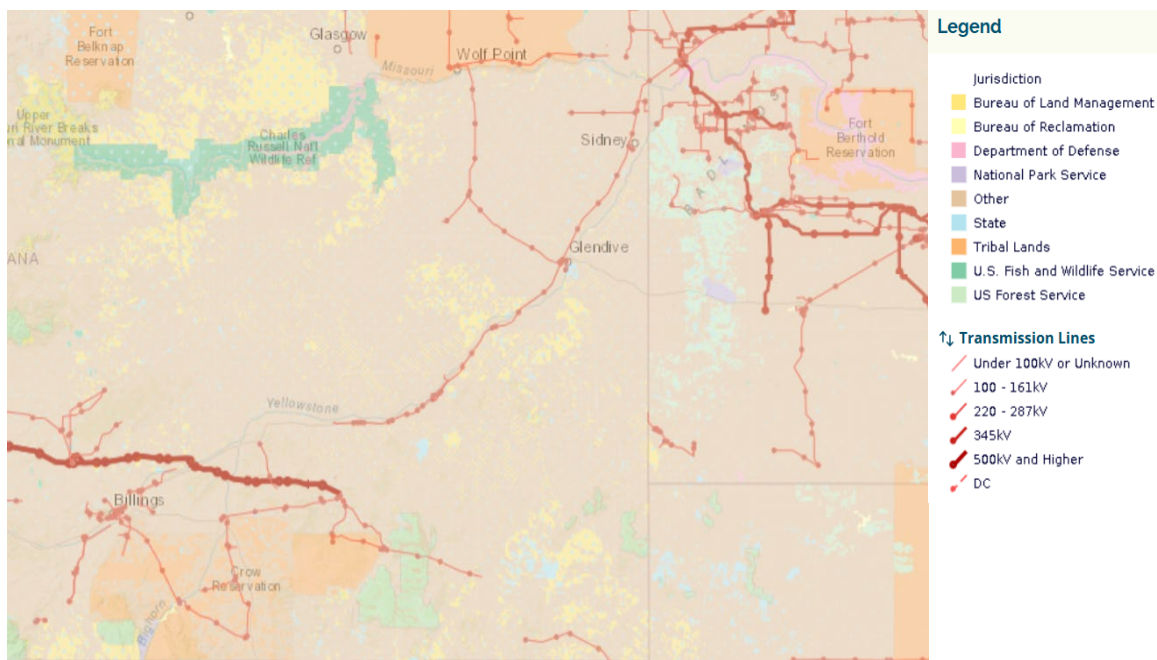
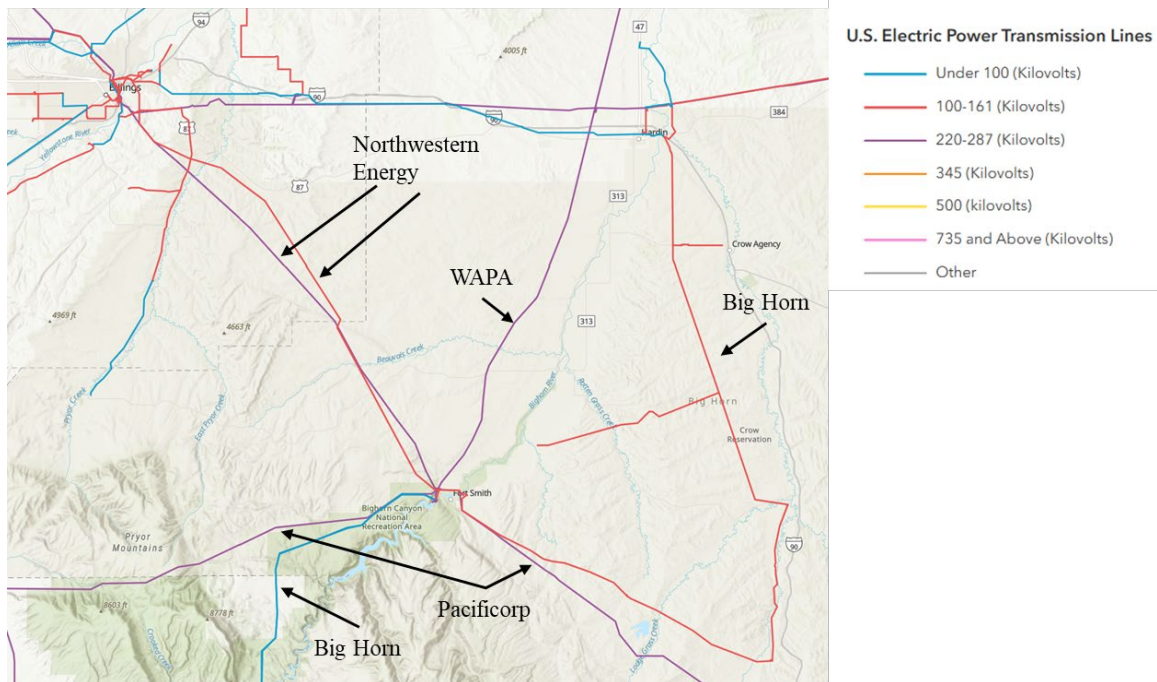
The other steps and processes of transmission siting need to be completed by the developers, in our case NTEA and the tribal communities, as they require direct interaction with federal, state, and local agencies for obtaining the necessary approvals.

Alternatives for Site Considerations

Preliminary analysis of the *Montana Chiefs’ Trail* indicates that constructing the proposed transmission corridor (Figure 2) might not be the best approach to connect the Crow and Northern Cheyenne Reservations (Montana) with the Fort Berthold Reservation (North Dakota). As seen in **Figure 10**, a more feasible alternative would be to connect them through existing transmission infrastructure. For instance, the Crow Reservation could be connected to the bulk power system using any of the following transmission lines:

1. 115 kV transmission line managed by Big Horn County Electric Cooperative
2. 69 kV transmission line managed by Big Horn County Electric Cooperative
3. 230 kV transmission line managed by PacifiCorp
4. 230 kV transmission line managed by the Western Area Power Administration
5. 230 kV transmission line managed by Northwestern Energy LLC
6. 115 kV transmission line managed by Northwestern Energy LLC

Regardless of the selected transmission line(s) to connect to, a transmission siting study is needed to analyze the path for interconnecting the generation resource(s) or load center(s) managed by the Native American Reservation(s) and the existing bulk power system, and to determine all approvals necessary. **Figure 11** shows some of the existing transmission lines that pass through surrounding federal lands and the agencies that have jurisdiction over those lands. All these different alternatives need to be evaluated through the SOW proposed.



Alternatives for Project Funding

During the research stage of preparing this document, a few alternatives were found as potential funding sources for the *Montana Chiefs' Trail* and/or other segments of the *Chiefs' Trail* transmission corridor.

Permitting and Siting Assistance Programs

In 2024, the DOE Grid Deployment Office announced two programs to advance critical transmission projects by accelerating siting and permitting processes [17]:

- ***Transmission Siting and Economic Development Grants Program***: This program provided up to \$371 million to 20 selected projects across 16 states to accelerate permitting of high-voltage transmission projects and support community infrastructure projects along major new and upgraded transmission lines.
- ***Transmission Acceleration Grants Program***: This program offered \$10 million to states, tribes, and non-profits to build capacity for transmission deployment and streamline transmission siting and permitting processes.

Public-Private Partnership Programs

The *Bipartisan Infrastructure Law* gave the DOE flexibility to partner with private companies on transmission projects, making it easier to develop transmission infrastructure in areas where local regulations could create obstacles. The DOE's Power Marketing Authorities (i.e., the Southwestern Power Administration and the Western Area Power Administration) have the authority to form partnerships for transmission development projects within their regions.

Transmission Facilitation Programs

The *Federal Power Act* authorizes the United States Secretary of Energy to designate certain areas ***National Interest Electric Transmission Corridors***, helping expedite transmission projects in regions where consumers are frequently harmed from the lack of transmission infrastructure, which directly contributes to higher electricity prices, frequent power outages from extreme weather, and longer outages. The designation of national corridors can streamline permitting and transmission siting projects, making it easier for companies to develop new or upgrade existing transmission infrastructure. In December 2024, the DOE announced that it is moving forward with three potential National Interest Electric Transmission Corridors, including the *Tribal Energy Access Corridor* (part of North Dakota, South Dakota, Nebraska, and 5 Tribal Reservations) that partially overlaps with the proposed *Chiefs' Trail* transmission corridor [18].

Through the *Infrastructure Investment and Jobs Act*, the DOE has created a \$2.5 billion revolving funds program called the ***Transmission Facilitation Program*** [19] to support the development of new interregional transmission lines across the United States. The program aims to help overcome financial obstacles in building large-scale transmission projects, upgrading existing infrastructure, and connecting microgrids in select areas.

CONCLUSION

The complex process of transmission siting involves several key steps and considerations that should be closely followed to ensure that a proposed project is completed successfully and timely:

First, the need for the new transmission line(s) must be identified, and a preliminary route corridor—which considers land use planning and pre-existing land restrictions—must be developed.

Second, a comprehensive review of the proposed route corridor—using advanced software technologies, such as GIS and power flow programs—must be conducted. This review should include (power) system planning studies and siting studies, which analyze electrical impacts on the existing bulk power system upon connection, and address potential impacts on biological resources, cultural resources, and other sensitive areas—these studies should be done in an iterative manner to guarantee the optimal selection of the transmission line route(s) considering all relevant factors.

Third, once the initial review has been completed, a more detailed evaluation of the proposed transmission route—using on-site investigations and consultations with relevant federal, state, and local government agencies—must be conducted. This evaluation should identify any potential land use restrictions or site-specific resource protection concerns that may impact the proposed project.

Last, all necessary permits and approvals from regulatory agencies must be obtained before the construction can begin. This includes complying with federal environmental regulations and state laws protecting sensitive areas and resources.

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